GATE Questions

1. **Graphs**

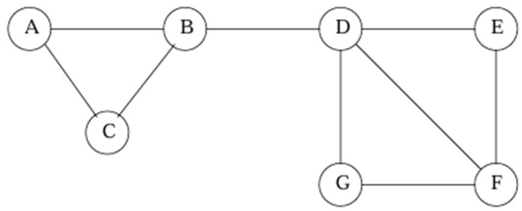


Figure 10.1: Example of a small social network

a. Consider the network in Figure 10.1. Please find the following

i. Degree distribution

ii. Diameter

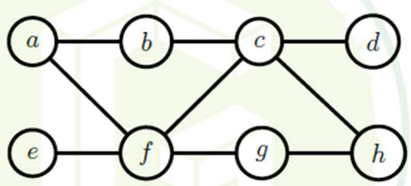
iii. Count all cliques

iv. Page rank of nodes

v. Begin with node C and D with labels Red and Green respectively. Perform label propagation and detect communities. Compute the modularity metric.

GATE Questions

Let G be a simple, unweighted, and undirected graph. A subset of the vertices and edges of G are shown below.



It is given that $a - b - c - d$ is a shortest path between a and d; $e - f - g - h$ is a shortest path between e and h; $a - f - c - h$ is a shortest path between a and h. Which of the following is/are NOT the edges of G?

(A) (b, d)

(B) (b, g)

(C) (b, h)

(D) (e, g)

Consider a directed graph $G = (V, E)$, where $V = \{0, 1, 2, \ldots, 100\}$ and $E = \{(i, j) : 0 < j - i \leq 2, \text{for all } i, j \in V\}$. Suppose the adjacency list of each vertex is in *decreasing* order of vertex number, and depth-first search (DFS) is performed at vertex 0. The number of vertices that will be discovered after vertex 50 is ____________(Answer in integer)

Q.52	Let $H, I, L,$ and N represent height, number of internal nodes, number of leaf nodes, and the total number of nodes respectively in a rooted binary tree. Which of the following statements is/are always **TRUE**?
(A)	$L \leq I + 1$
(B)	$H + 1 \leq N \leq 2^{H+1} - 1$
(C)	$H \leq I \leq 2^{H} - 1$
(D)	$H \leq L \leq 2^{H-1}$

Q.61	Let $\boldsymbol{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$, and let $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5$ be the singular values of the matrix $\boldsymbol{M} = \boldsymbol{u}\boldsymbol{u}^{\boldsymbol{T}}$ (where $\boldsymbol{u}^{\boldsymbol{T}}$ is the transpose of $\boldsymbol{u}$). The value of $\sum_{i=1}^{5} \sigma_i$ is ______.

2. Linear Algebra

Book questions

1. Let A be

$$A = \begin{pmatrix} 1 & 2 \\ -1 & 2 \\ 1 & -2 \\ -1 & -2 \end{pmatrix}$$

   1. *Run the power method starting from* $x = \binom{1}{1}$ *for* $k = 3$ *steps. What does this give as an estimate of* v_1*?*
   2. *What actually are the* v_i*'s,* σ_i*'s, and* u_i*'s? It may be easiest to do this by computing the eigenvectors of* $B = A^T A$.
   3. *Suppose matrix* A *is a database of restaurant ratings: each row is a person, each column is a restaurant, and* a_{ij} *represents how much person* i *likes restaurant* j*. What might* v_1 *represent? What about* u_1*? How about the gap* $\sigma_1 - \sigma_2$*?*

2. Let A be a square n $\vee$n matrix whose rows are orthonormal. Prove that the columns of A are orthonormal.

3. **Exercise 3.8** *Suppose* A *is a* $n \times n$ *matrix with block diagonal structure with* k *equal size blocks where all entries of the* i^{th} *block are* a_i *with* $a_1 > a_2 > \cdots > a_k > 0$*. Show that* A *has exactly* k *nonzero singular vectors* $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_k}$ *where* $\mathbf{v_i}$ *has the value* $(\frac{k}{n})^{1/2}$ *in the coordinates corresponding to the* i^{th} *block and 0 elsewhere. In other words, the singular vectors exactly identify the blocks of the diagonal. What happens if* $a_1 = a_2 = \cdots = a_k$*? In the case where the* a_i *are equal, what is the structure of the set of all possible singular vectors?*

4.

Exercise 3.10 *Verify that the sum of* r*-rank one matrices* $\sum_{i=1}^{r} c_i \mathbf{x_i} \mathbf{y_i}^T$ *can be written as* XCY^T*, where the* $\mathbf{x_i}$ *are the columns of* X*, the* $\mathbf{y_i}$ *are the columns of* Y*, and* C *is a diagonal matrix with the constants* c_i *on the diagonal.*

Exercise 3.11 *Let* $\sum_{i=1}^{r} \sigma_i \mathbf{u_i} \mathbf{v_i}^T$ *be the SVD of* A*. Show that* $\left|\mathbf{u_1}^T A\right| = \sigma_1$ *and that* $\left|\mathbf{u_1}^T A\right| = \max_{|\mathbf{u}|=1} \left|\mathbf{u^T} A\right|$.

Exercise 3.12 *If $\sigma_1, \sigma_2, \ldots, \sigma_r$ are the singular values of A and $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$ are the corresponding right-singular vectors, show that*

1. $A^T A = \sum_{i=1}^{r} \sigma_i^2 \mathbf{v_i} \mathbf{v_i}^T$
2. $\mathbf{v}_1, \mathbf{v}_2, \ldots \mathbf{v}_r$ *are eigenvectors of* $A^T A$.
3. *Assuming that the eigenvectors of $A^T A$ are unique up to multiplicative constants, conclude that the singular vectors of A (which by definition must be unit length) are unique up to sign.*

Exercise 3.13 *Let $\sum_i \sigma_i u_i v_i^T$ be the singular value decomposition of a rank r matrix A. Let $A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T$ be a rank k approximation to A for some $k < r$. Express the following quantities in terms of the singular values $\{\sigma_i, 1 \leq i \leq r\}$.*

1. $||A_k||_F^2$
2. $||A_k||_2^2$
3. $||A - A_k||_F^2$
4. $||A - A_k||_2^2$

Exercise 3.14 *If A is a symmetric matrix with distinct singular values, show that the left and right singular vectors are the same and that $A = VDV^T$.*

Exercise 3.15 *Let A be a matrix. How would you compute*

$$\mathbf{v_1} = \arg\max_{|\mathbf{v}|=1} |A\mathbf{v}|?$$

How would you use or modify your algorithm for finding $\mathbf{v_1}$ to compute the first few singular vectors of A.

Exercise 3.16 *Use the power method to compute the singular value decomposition of the matrix*

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Exercise 3.17

1. *Write a program to implement the power method for computing the first singular vector of a matrix. Apply your program to the matrix*

$$A = \begin{pmatrix} 1 & 2 & 3 & \cdots & 9 & 10 \\ 2 & 3 & 4 & \cdots & 10 & 0 \\ \vdots & \vdots & \vdots & & & \vdots \\ 9 & 10 & 0 & \cdots & 0 & 0 \\ 10 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Exercise 3.19 *Prove that the eigenvalues of a symmetric real valued matrix are real.*

Exercise 3.20 *Suppose A is a square invertible matrix and the SVD of A is $A = \sum_i \sigma_i u_i v_i^T$. Prove that the inverse of A is $\sum_i \frac{1}{\sigma_i} v_i u_i^T$.*

Exercise 3.21 *Suppose A is square, but not necessarily invertible and has SVD $A = \sum_{i=1}^{r} \sigma_i u_i v_i^T$. Let $B = \sum_{i=1}^{r} \frac{1}{\sigma_i} v_i u_i^T$. Show that $B A \mathbf{x} = \mathbf{x}$ for all $\mathbf{x}$ in the span of the right-singular vectors of A. For this reason B is sometimes called the pseudo inverse of A and can play the role of A^{-1} in many applications.*

Exercise 3.22

1. *For any matrix A, show that $\sigma_k \leq \frac{||A||_F}{\sqrt{k}}$.*

2. *Prove that there exists a matrix B of rank at most k such that $||A - B||_2 \leq \frac{||A||_F}{\sqrt{k}}$.*

3. *Can the 2-norm on the left hand side in (2) be replaced by Frobenius norm?*

GATE questions

Let $D = \{x^{(1)}, \ldots, x^{(n)}\}$ be a dataset of n observations where each $x^{(i)} \in \mathbb{R}^{100}$. It is given that $\sum_{i=1}^{n} x^{(i)} = 0$. The covariance matrix computed from D has eigenvalues $\lambda_i = 100^{2-i}$, $1 \leq i \leq 100$. Let $u \in \mathbb{R}^{100}$ be the direction of maximum variance with $u^\top u = 1$.

The value of

$$\frac{1}{n}\sum_{i=1}^{n} \left(u^\top x^{(i)}\right)^2 = \_\_\_\_\_\_\_\_\_\_$$

(Answer in integer)

An $n \times n$ matrix A with real entries satisfies the property: $\|Ax\|^2 = \|x\|^2$, for all $x \in \mathbb{R}^n$, where $\|\cdot\|$ denotes the Euclidean norm. Which of the following statements is/are ALWAYS correct?

(A) A must be orthogonal

(B) $A = I$, where I denotes the identity matrix, is the only solution

(C) The eigenvalues of A are either $+1$ or -1

(D) A has full rank

Let x_1, x_2, x_3, x_4, x_5 be a system of orthonormal vectors in $\mathbb{R}^{10}$. Consider the matrix $A = x_1 x_1^\top + \ldots + x_5 x_5^\top$. Which of the following statements is/are correct?

(A) Singular values of A are also its eigenvalues

(B) Singular values of A are either 0 or 1

(C) Determinant of A is 1

(D) A is invertible

Let $A \in \mathbb{R}^{n \times n}$ be such that $A^3 = A$. Which one of the following statements is ALWAYS correct?

(A) A is invertible

(B) Determinant of A is 0

(C) The sum of the diagonal elements of A is 1

(D) A and A^2 have the same rank

Let $\{x_1, x_2, \ldots, x_n\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. Let the (i,j)-th element of matrix $A \in \mathbb{R}^{n \times n}$ be given by $A_{ij} = x_i^\top x_j$, $1 \leq i, j \leq n$. Which one of the following statements is correct?

(A) A is invertible

(B) 0 is a singular value of A

(C) Determinant of A is 0

(D) $z^\top A z = 0$ for some non-zero $z \in \mathbb{R}^n$

Let $A = I_n + x x^\top$, where I_n is the $n \times n$ identity matrix and $x \in \mathbb{R}^n$, $x^\top x = 1$. Which of the following options is/are correct?

(A) Rank of A is n

(B) A is invertible

(C) 0 is an eigenvalue of A

(D) A^{-1} has a negative eigenvalue

Which of the following statements is/are correct?

(A) $\mathbb{R}^n$ has a unique set of orthonormal basis vectors

(B) $\mathbb{R}^n$ does not have a unique set of orthonormal basis vectors

(C) Linearly independent vectors in $\mathbb{R}^n$ are orthonormal

(D) Orthonormal vectors $\mathbb{R}^n$ are linearly independent

Consider the following Python declarations of two lists.

```
A=[1,2,3]
B=[4,5,6]
```

Which one of the following statements results in `A`= $[1, 2, 3, 4, 5, 6]$?

(A) `A.extend(B)`

(B) `A.append(B)`

(C) `A.update(B)`

(D) `A.insert(B)`

The sum of the elements in each row of $A \in \mathbb{R}^{n\times n}$ is 1. If $B = A^3 - 2A^2 + A$, which one of the following statements is correct (for $x \in \mathbb{R}^n$)?

(A) The equation $Bx = 0$ has no solution

(B) The equation $Bx = 0$ has exactly two solutions

(C) The equation $Bx = 0$ has infinitely many solutions

(D) The equation $Bx = 0$ has a unique solution

The number of additions and multiplications involved in performing Gaussian elimination on any $n \times n$ upper triangular matrix is of the order

(A) $O(n)$

(B) $O(n^2)$

(C) $O(n^3)$

(D) $O(n^4)$

Q.48	Which of the following statements is/are **TRUE**? **Note**: $\mathbb{R}$ denotes the set of real numbers.
(A)	There exist $\boldsymbol{M} \in \mathbb{R}^{3\times 3}, \boldsymbol{p} \in \mathbb{R}^3$, and $\boldsymbol{q} \in \mathbb{R}^3$ such that $\boldsymbol{M}\mathbf{x} = \boldsymbol{p}$ has a unique solution and $\boldsymbol{M}\mathbf{x} = \boldsymbol{q}$ has infinite solutions.
(B)	There exist $\boldsymbol{M} \in \mathbb{R}^{3\times 3}, \boldsymbol{p} \in \mathbb{R}^3$, and $\boldsymbol{q} \in \mathbb{R}^3$ such that $\boldsymbol{M}\mathbf{x} = \boldsymbol{p}$ has no solutions and $\boldsymbol{M}\mathbf{x} = \boldsymbol{q}$ has infinite solutions.
(C)	There exist $\boldsymbol{M} \in \mathbb{R}^{2\times 3}, \boldsymbol{p} \in \mathbb{R}^2$, and $\boldsymbol{q} \in \mathbb{R}^2$ such that $\boldsymbol{M}\mathbf{x} = \boldsymbol{p}$ has a unique solution and $\boldsymbol{M}\mathbf{x} = \boldsymbol{q}$ has infinite solutions.
(D)	There exist $\boldsymbol{M} \in \mathbb{R}^{3\times 2}, \boldsymbol{p} \in \mathbb{R}^3$, and $\boldsymbol{q} \in \mathbb{R}^3$ such that $\boldsymbol{M}\mathbf{x} = \boldsymbol{p}$ has a unique solution and $\boldsymbol{M}\mathbf{x} = \boldsymbol{q}$ has no solutions.

Q.47	Select all choices that are subspaces of $\mathbb{R}^3$. **Note**: $\mathbb{R}$ denotes the set of real numbers.
(A)	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \ \mathbf{x} = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \alpha, \beta \in \mathbb{R}\right\}$
(B)	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \ \mathbf{x} = \alpha^2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \beta^2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \alpha, \beta \in \mathbb{R}\right\}$
(C)	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \ 5x_1 + 2x_3 = 0, 4x_1 - 2x_2 + 3x_3 = 0\right\}$
(D)	$\left\{\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \ 5x_1 + 2x_3 + 4 = 0\right\}$

Q.49	Let $\mathbb{R}$ be the set of real numbers, U be a subspace of $\mathbb{R}^3$ and $\boldsymbol{M} \in \mathbb{R}^{3\times 3}$ be the matrix corresponding to the projection on to the subspace U. Which of the following statements is/are **TRUE**?
(A)	If U is a 1-dimensional subspace of $\mathbb{R}^3$, then the null space of $\boldsymbol{M}$ is a 1-dimensional subspace.
(B)	If U is a 2-dimensional subspace of $\mathbb{R}^3$, then the null space of $\boldsymbol{M}$ is a 1-dimensional subspace.
(C)	$\boldsymbol{M}^2 = \boldsymbol{M}$
(D)	$\boldsymbol{M}^3 = \boldsymbol{M}$

3. Python

Consider the following Python code snippet.

```
def f(a,b):
    if (a==0):
        return b
    if (a%2==1):
        return 2*f((a-1)/2,b)
    return b+f(a-1,b)
print(f(15,10))
```

The value printed by the code snippet is ____________(Answer in integer)

```
Create empty stack S

Set x=0, flag=0, sum=0

Push x onto S

while (S is not empty){

  if (flag equals 0){

    Set x = x+1

    Push x onto S}

  if (x equals 8):

    Set flag=1

  if (flag equals 1){

    x = Pop(S)

    if (x is odd):

      Pop(S)

    Set sum = sum + x}

 }

 Output sum
```

The value of `sum` output by a program executing the above pseudocode is ____________

(Answer in integer)

Q.38	Consider the following Python code:

```
def count(child_dict, i):
    if i not in child_dict.keys():
         return 1
    ans = 1
    for j in child_dict[i]:
        ans += count(child_dict, j)
    return ans

child_dict = dict()
child_dict[0] = [1,2]
child_dict[1] = [3,4,5]
child_dict[2] = [6,7,8]
print(count(child_dict,0))
```

Which **ONE** of the following is the output of this code?

(A)	6
(B)	1
(C)	8
(D)	9

Q.39	Consider the function **computeS** (X) whose pseudocode is given below:
	
	<u>computeS</u> <u>(X)</u>
	$S[1] \leftarrow 1$
	for $i \leftarrow 2$ to $length(X)$
	  $S[i] \leftarrow 1$
	  if $X[i-1] \leq X[i]$
	      $S[i] \leftarrow S[i] + S[i-1]$
	  end if
	end for
	return S
	
	Which **ONE** of the following values is returned by the function **computeS** (X) for $X = [6, 3, 5, 4, 10]$?

(A)	[1, 1, 2, 3, 4]
(B)	[1, 1, 2, 3, 3]
(C)	[1, 1, 2, 1, 2]
(D)	[1, 1, 2, 1, 5]

Consider the following Python code snippet.

```
A={"this","that"}

B={"that","other"}

C={"other","this"}

while "other" in C:

    if "this" in A:

      A,B,C=A-B,B-C,C-A

    if "that" in B:

      A,B,C=C|A,A|B,B|C
```

When the above program is executed, at the end, which of the following sets contains `"this"`?

(A) Only `A`

(B) Only `B`

(C) Only `C`

(D) `A`, `C`